

ConectaMapa: from hidden broadband demand to rollout decisions

Regional internet providers in Mexico decide where to extend fixed broadband **without seeing real demand**. ConectaMapa lets households and small businesses register unmet demand and helps ISPs decide **which locality to connect next, and at what price**.

1. Problem

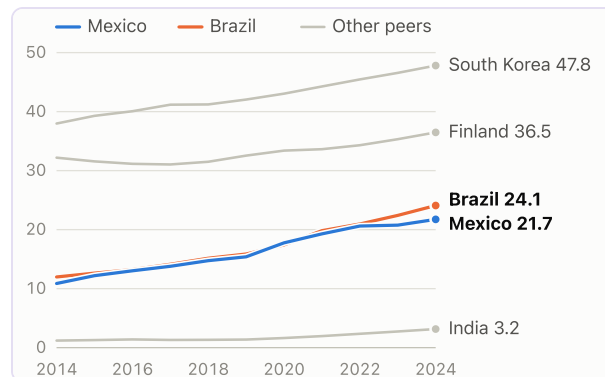
Actor: the expansion manager of a regional ISP (small fibre or fixed-wireless operator). **Decision:** which locality to build next quarter and at what monthly price. **Phenomenon:** unmet fixed-broadband demand at locality level, invisible to providers. **Consequence:** viable localities stay unconnected and ISPs build where take-up disappoints.

2. Evidence

- Mexico: **21.7 fixed broadband subscriptions per 100 people** (2024), 4th of 5, under half of South Korea (47.8). Since 2021 Brazil added +4.3 points vs Mexico +2.4.
- ≈**8.6 million** households (21.7%) had no internet at home in 2025, from 90.5% connected in Mexico City to 53.9% in Chiapas (INEGI, ENDUTIH 2025). **The gap is geographic.**
- In Brazil, small regional providers hold **63.3% of fixed broadband** (Anatel, end-2025): a documented channel for reaching underserved areas (an association, not proof of causation).

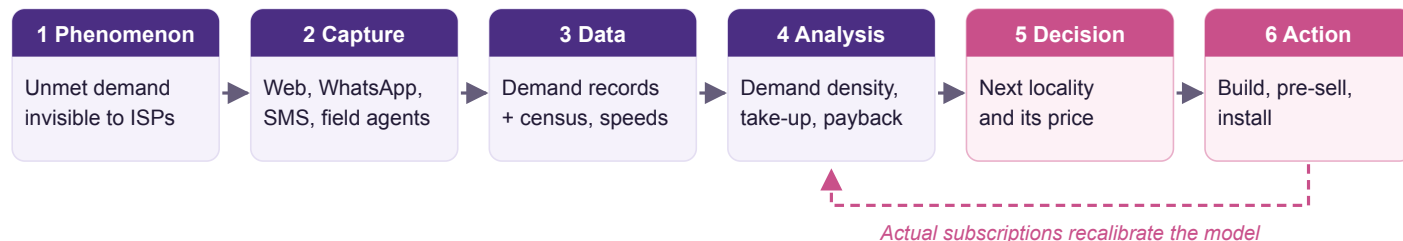
Since 2021, Brazil has pulled ahead of Mexico

Fixed broadband subscriptions per 100 people, 2014–2024



Source: ITU via World Bank WDI (IT.NET.BBND.P2), Part I raw data.

3. Digital transformation pipeline



The ranking shifts from "where many people live" to "where many people will subscribe": a small locality with high registered demand can move ahead of a large one.

4. Analytical strategy

Level	Question	Technique	Decision supported
Descriptive	How much unmet demand did each locality register?	Rates per 100 households, heat maps	Shortlist for field survey
Diagnostic	Why is demand unmet where a provider exists?	Segmentation, regression with controls	New network, cheaper plan or upgrade
Predictive	If we build in X, how many will subscribe in 6 months?	Classification models validated on past builds	Go / no-go per locality
Prescriptive	With budget B and crews, which localities and prices?	Integer optimization, scenarios	Quarterly rollout plan

A demand map shows *stated* interest; prediction must learn who actually pays; prescription adds constraints and an objective.

5. Patents and innovation (WIPO query)

Four digital fields (telecommunications, digital communication, computer technology, IT methods for management), by applicant's origin. **Mexico is last and falling:** ≈89 publications a year in 2021–2023, –21% vs 2014–2016; South Korea publishes ≈1,500× more per person. Patents protect **novel, inventive, industrially applicable** technical solutions (LFPPI Art. 48); business methods and software *as such* are excluded (Art. 47). ConectaMapa's edge is **data and network effects, not IP**. A possible inventive component is a method that fuses speed tests, census geodata and network-design parameters to estimate a locality's viability. Counts measure invention attempts, not adoption or success.

Origin	Per year	Per million people
South Korea	54,245	1,048.9
Finland	3,941	708.7
India	5,501	3.9
Brazil	315	1.5
Mexico	89	0.7

2021–2023 average, four fields (2024 incomplete). WIPO IP Statistics, consulted 29 Sep 2026; population: World Bank. Evidence: part2/wipo/.

6. Business model

- **Value creation:** households get visibility and a connection; ISPs get validated demand, lower risk and cheaper acquisition; municipalities get evidence for co-investment.
- **Value delivery:** free registration via web, WhatsApp or SMS (plus field agents); dashboard with demand map and forecasts for ISPs; aggregated reports for government.
- **Value capture:** ISP **SaaS** subscription + **success fee** per activated subscriber + **aggregated, anonymized data** reports.

Model: marketplace + SaaS + data-driven. The price-sensitive side (households) is free; the side that benefits and can pay (ISPs) pays. We do not use advertising, because it would erode trust and privacy.

8. Scalability: what if users grow 10×?

Grows slowly: cloud compute and storage, software, model training. **Keeps growing:** SMS/WhatsApp messages, field agents in hard-to-reach areas, support, fraud verification, B2B sales, data-protection compliance. The digital layer scales; **the physical networks ISPs must build do not**. Scalability is low marginal cost, not zero cost.

9. Data quality and responsibility

- **Quality:** self-reported willingness to pay is inflated, and speed tests depend on the device → take-up is overestimated.
- **Representativeness:** households without internet are the least able to register online (rural internet use is 75.2% vs 88.9% urban), so offline channels are required or the platform reproduces the divide.
- **Privacy:** home location + phone + income signals → consent, locality-level aggregates, retention limits.
- **Bias:** a model trained on past (denser, richer) builds under-predicts rural take-up → fairness constraints and a minimum share for high-marginalization localities.
- **Correlation ≠ causation:** localities with more registrations show higher take-up, but income or community organization may drive both. Testing needs comparable localities with and without campaigns.

10. Digital is material · AI and energy

Physical elements: household phones; cloud servers with cooling; fibre, optical terminals or radios and towers; electricity and backup power. **Material cost:** fibre per km plus equipment per home. **Energy cost:** network equipment runs 24/7, and rural areas need batteries, solar or diesel. **Externality:** e-waste from replaced routers and emissions from diesel backup.

7. Platform and network effects

Multi-sided platform: households ↔ ISPs, with municipalities as co-funders. **Rules:** one verified registration per household; ISPs publish price, speed and date before seeing contacts; consent-based data sharing, no resale; post-install speed test and ratings.

- **Indirect, positive:** more households → more ISPs → more competition and coverage → more households.
- **Direct, positive (neighbours):** each registration pushes the locality toward its viability threshold ("demand pooling").
- **Direct, negative (ISPs):** competing ISPs in one locality lower each other's value.
- Effects are **local**, so the cold start is solved locality by locality.

One system, two directions: Mexico's electricity grid. AI helps: forecasts of demand and solar output let the grid operator (e.g., CENACE) keep fewer fossil plants in reserve; ISPs on the same grid put idle radios to sleep at night. **AI demands:** a typical AI data centre uses as much electricity as 100,000 households (IEA); new ones on Mexico's grid compete for the capacity rural networks need. We use a simple registration form, not an LLM chatbot.

11. Final proposal

Problem	Regional ISPs expand fixed broadband without seeing real demand; viable underserved localities stay unconnected.
Evidence	21.7 subscriptions per 100 people (4th of 5); slower growth than Brazil since 2021; ≈8.6 M households without internet; strong state-level gaps; regional ISPs hold 63.3% of Brazil's fixed broadband.
Strategy	Online and offline registration joined with census and speed data; descriptive → diagnostic → predictive → prescriptive analysis.
Decision	Which locality an ISP connects next and at what price; where municipalities co-invest.
Value	Households and small businesses (connection), ISPs (lower risk and acquisition cost), municipalities (evidence).
Business model	Free for households; SaaS + success fee for ISPs; aggregated reports for the public sector.
Limitations	ISPs lacking capital; permits and rights of way; fake demand; cold start; localities never viable without subsidy; privacy incidents; regulatory change.

Sources: ITU via World Bank WDI (IT.NET.BBND.P2, IT.NET.USER.ZS), Part I raw data · INEGI, ENDUTIH 2025 (Reporte de Resultados 19/26, 16 Jun 2026) · Anatel, Relatório de Monitoramento da Competição 4T2025 · IEA, Energy and AI (2025) · Ley Federal de Protección a la Propiedad Industrial (2020), Arts. 47–48 · WIPO IP Statistics Data Center, 4a Patent publications by technology (consulted 29 Sep 2026) · World Bank WDI, population (SP.POP.TOTL). Full analysis: **Annex** (Team1_Part2_Annex.pdf) · traceability: part2/evidence_log.csv.